# Akshay Borse

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## PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years building distributed systems, security services, and production ML systems at Microsoft and Amazon. Experience spans 237K+ AI agents, 100K+ TPS, and commercial and US government cloud deployments.

## EXPERIENCE

**Senior Software Engineer | Microsoft | Redmond, WA | Sep 2025 to Present**

**Enterprise AI Platforms • AI Governance • Sovereign Cloud** • C# • .NET • Azure • Cosmos DB • Event Hub • Microsoft Purview • Kusto • Bicep

- Built tenant-configurable detection and risk evaluation for agentic and non-agentic AI in Microsoft Purview Insider Risk Management, covering 237K+ agents across 13K organizations.
- **Reconstructed 90 days of audit history across 40+ global forests**, allowing immediate risk evaluation for newly onboarded AI agents.
- Led architecture and global rollout of Risk Detection, Adaptive Protection, and AI governance infrastructure across Commercial, **GCC, GCCH, and DoD sovereign cloud**, ensuring region-specific compliance while maintaining feature parity across commercial and government tenants.
- Engineered a regionally isolated event-driven AI-agent ingestion pipeline spanning 28 Azure regions, processing **135K+ security signals daily** while preventing AI workloads from impacting human-user processing pipelines.
- Integrated Microsoft Graph, Entra, Defender, Sentinel, and Purview signals into a cross-domain correlation system, combining identity, behavioral, security, and compliance telemetry to detect risky AI agents and support policy enforcement across enterprise environments.

**Software Engineer 2 | Amazon | Seattle, WA | Aug 2019 – Aug 2025**

**Platform Engineering • Distributed Systems • AI Infrastructure • Developer Productivity** • Java • Python • AWS • Docker • ECS • Lambda • DynamoDB • Step Functions • REST APIs • gRPC • Microservices • Kinesis • OpenSearch • CloudWatch • Bedrock • SageMaker • RAG • XGBoost

- Created a typed DSL and one-click provisioning pipeline adopted by 3 teams and 40 engineers to bootstrap 50+ AWS services, saving an estimated 40–50 engineering weeks.
- Designed and implemented the zero-downtime consolidation of nine ingestion pipelines, five dependency routers, and three routing services into reusable multi-service pipelines, using weighted routing and idempotent queue handling to simplify maintenance and fleet-wide upgrades.
- Built a stateful AWS Step Functions service that replayed 200K+ order events daily across three regions, re-evaluating canceled orders for 30 days and increasing restocked-item sales by 13%.
- Designed and delivered a reusable API library for runtime Log4j configuration, with audit logging and automatic 30-minute rollback, enabling integration across services and helping 12 engineers diagnose production incidents without 1+ hour redeployments.
- Led development of a real-time XGBoost delivery-risk system on SageMaker that replaced static delay rules and informed automated recovery, early customer notifications, and alternate-seller recommendations, contributing to a 7% increase in global sales.
- Launched an LLM-assisted localization system using Amazon Bedrock, RAG, and a DynamoDB catalog of approved content to personalize compliance-reviewed emails across 37 templates and 28 locales.
- Owned the API, SageMaker infrastructure, deployment, monitoring, and rollout for real-time quantile-regression models that generated delivery-date ranges with rule-based fallbacks, contributing an estimated $28M in annualized impact measured through A/B testing.
- Proposed and built a **dual-protocol orchestrator hosting REST and gRPC servers** with shared business logic, avoiding translation overhead while coordinating 16 AWS microservices at 100K+ TPS during a large-scale monolith migration.
- Coordinated Prime Day and Black Friday readiness for 60+ services by translating traffic forecasts and historical peaks into load tests, capacity targets, and dependency-throttling fixes; the fleet recorded zero major event-related incidents.
- Led an order-upgrade service using inventory signals and confidence thresholds, with false-positive safeguards and rollback handling, reducing unnecessary customer reorders and increasing early deliveries by 16%.
- Conceived and delivered an RBAC-controlled self-service tool enabling PMs and TPMs to manually update customer delivery promises when inventory signals were delayed, **saving an estimated 100–300 engineering hours annually and reducing on-call workload**.
- Identified and standardized Docker build optimizations across 40+ CI/CD pipelines using multi-stage builds, dependency pruning, smaller base images, and improved layer caching, **reducing image size by 83% (1.4GB), build time by 50% (90 seconds), and storage by 60GB+**.
- Cut CloudWatch costs by $250K+ per month by introducing logging standards and a shared library for retention, log levels, sampling, filtering, and aggregation without reducing production visibility.
- Initiated and led post-on-call reviews across 40+ services and 15 engineers, driving alert tuning, automation, retry, safeguards, architectural fixes, and runbooks that reduced Sev2 incidents by 40% over two months while cutting pages and mitigation effort.
- Expanded the service-creation library with unit, integration, load, stress, canary, and fault-injection test workflows, plus metric-based rollback for dependency failures, latency, and throttling.

**Software Engineering Intern | Liquiron | San Jose, CA | Dec 2018 to Jan 2019**

**Backend Development • Serverless Architecture • Full-Stack Development** • Node.js • Vue.js • Firebase • OAuth 2.0 • Passport.js • REST APIs • Jest

- Re-architected authentication using OAuth 2.0 and Passport.js, improving security, modularity, and maintainability.
- Migrated backend services to Firebase serverless infra & developed REST APIs supporting SPA workflows, reducing server load by 36%.

**Software Engineer | Persistent Systems | Pune, India | Sep 2016 to Jul 2017**

**Machine Learning • Full-Stack Development • Web Modernization** • Python • JavaScript • Node.js • REST APIs • CNN • AWS EC2

- Developed a real-time CNN-based sentiment classification system achieving 87% accuracy across seven sentiment categories.
- Modernized legacy web applications into a single-page architecture, improving response time by 25% and offline performance by 9%, while introducing an engineering feedback system adopted by six development teams.

## EDUCATION

- Santa Clara University, Santa Clara, CA | Master’s in Computer Science and Engineering. | Sep 2017 to Jun 2019 | GPA: 3.77
- Pune Institute of Computer Technology (PICT), India | Bachelor’s in Computer Science and Engineering. | Jun 2012 to Jun 2016 | GPA: 3.34

## FEATURED PROJECTS -

• C • C++ • TypeScript • React • Node.js • Kafka • Redis • Neo4j • WebSockets • Progressive Web Apps • Offline-First Architecture • Vitest • Playwright • Linux Kernel Development • F2FS • Filesystem Performance • Developer Tooling • Repository Analysis • Workflow Automation

- **LeetDesign:** Built an interactive system-design simulator for modeling scalability, fault tolerance, and architectural trade-offs through executable service and workload models.
- **FinanceOS:** Transformed financial transactions into explainable insights and personalized recommendations through FinanceOS, an AI-powered financial intelligence platform combining deterministic processing with LLM-assisted reasoning.
- **Arsenal:** Built a developer-tooling platform for project scaffolding, workflow automation, and context-efficient coding workflow
- **Orbit:** Built an offline-first life-management platform for organizing tasks, habits, recurring routines, and workload insights.
- **Ad-infinitum:** Linux filesystem research project exploring F2FS garbage-collection policies through kernel instrumentation, performance benchmarking, and latency optimization.

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